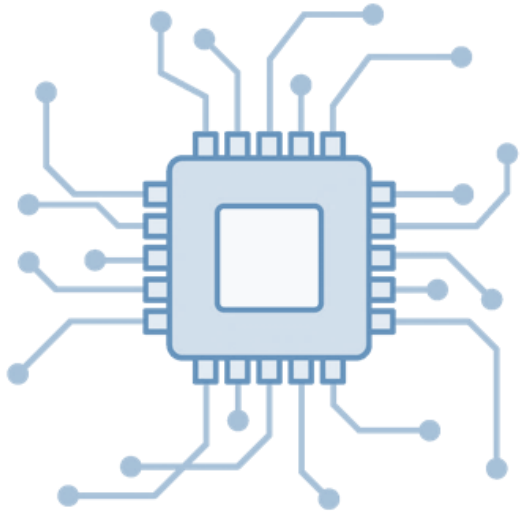
CodeWeek.eu

IEEE University of Porto - Student Branch

A Workshop on the Internet-of-Things

From the **Cloud** to the *Edge* and the *Mist*



João Pedro Dias

23 October 2020

CodeWeek.

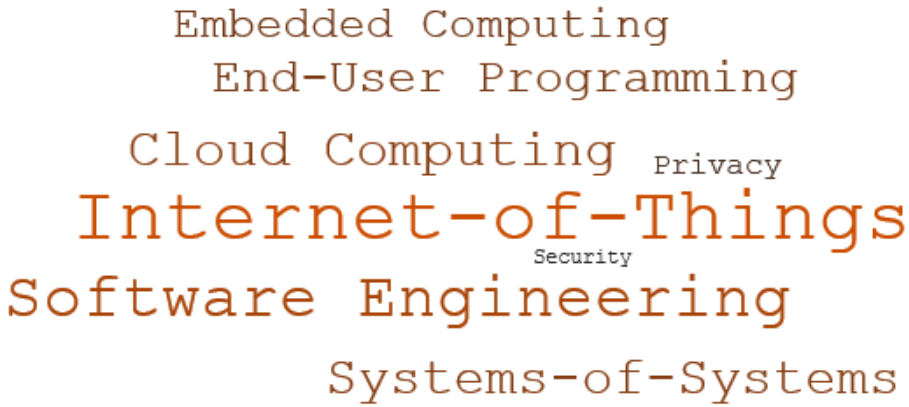
$ whoami

- From Porto, PT

- Researcher @ INESC TEC
- PhD Student @ FEUP/DEI

- Previous talks/workshops
  - PixelsCamp 2017 and 2019
  - 0xOPOSEC meetups
  - …

- Me around the web:
  - https://jpdias.me
  - https://twitter.com/jpd1as/
  - jpmdias@fe.up.pt



Internet-of-Things *by the standards*

- *"An infrastructure of **interconnected objects**, people, systems and information resources together with intelligent services to allow them to **process information of the physical and the virtual world and react**."*

ISO/IEC JTC 1 Internet of Things (IoT)

What *is* IoT *really*?

- *A network of physical objects — **things** — that are embedded with **sensors, actuators, software, and other** technologies for the purpose of **connecting and exchanging data** with other devices and systems **over the Internet.***

From Wikipedia, the free encyclopedia











nest





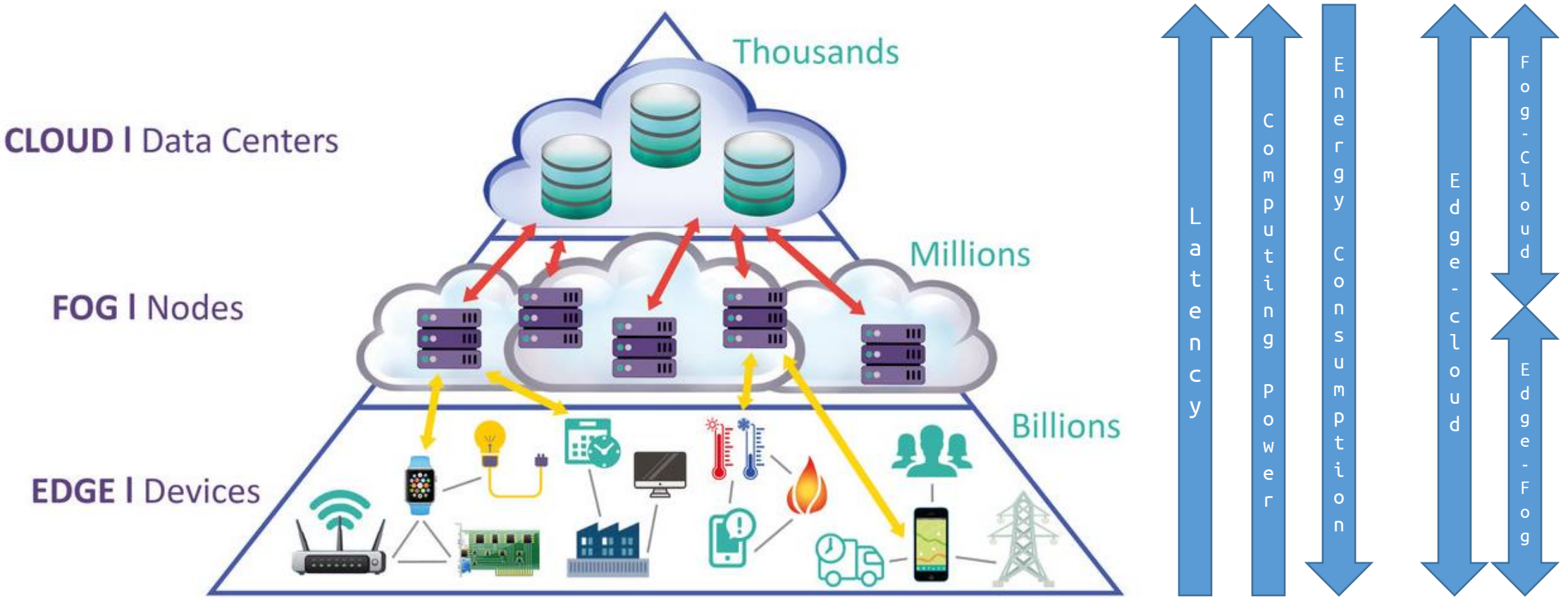
Google Home

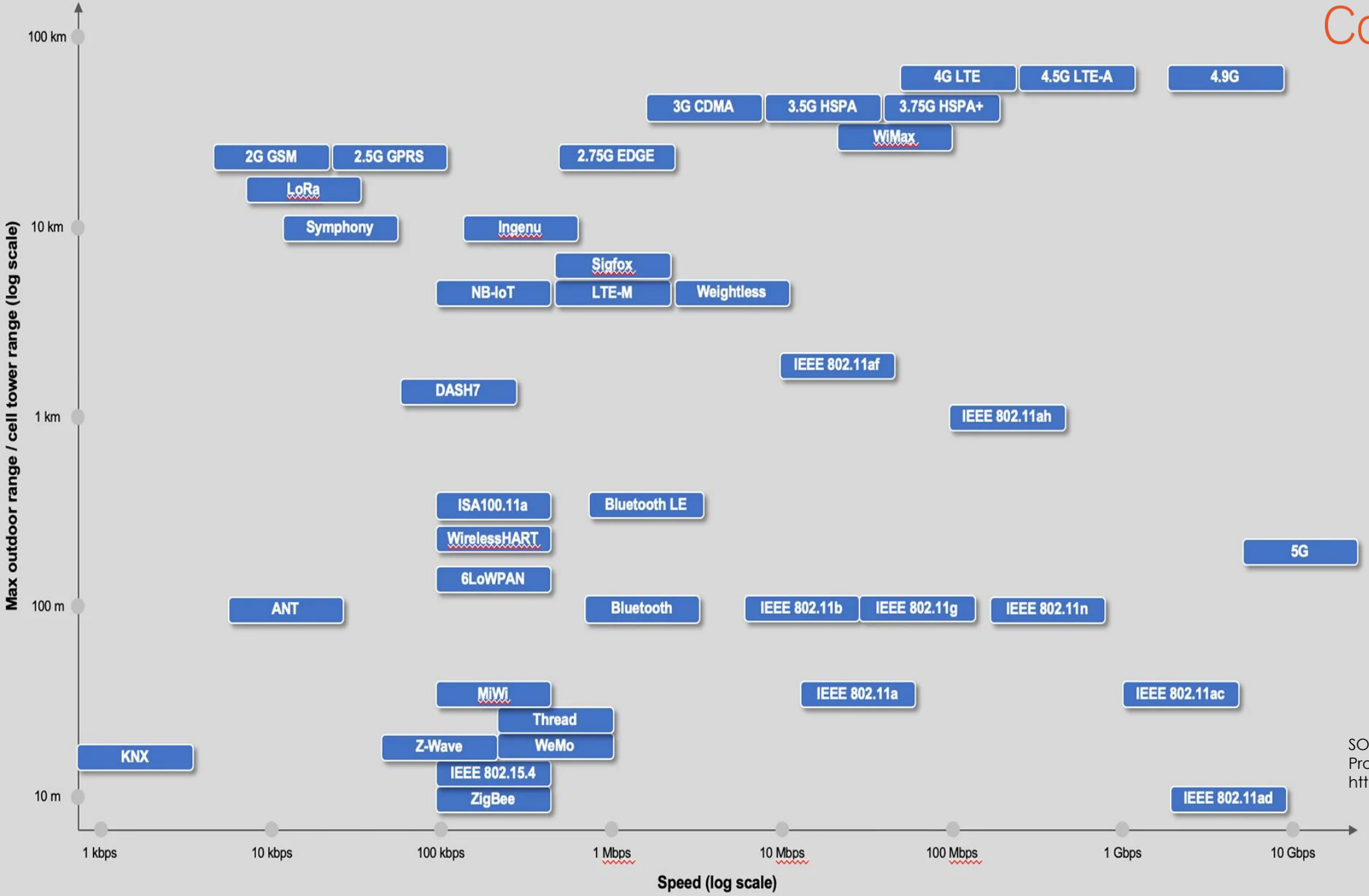
Hands-free smart speaker





Cloud, Fog and Mist (Edge)





SOURCE: Internet of Things (IoT) Wireless Protocols, https://5g.security/iot-wireless-protocols/

IKEA TRÅDFRI example



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The DIY path

Because buying things is expensive... and there's no fun in that.

Also, personalize your system, after all, it's yours.

Plan of attack

1. Define your architecture
   - Edge-Cloud, Edge-Fog-Cloud, Edge-Fog (local only)
2. Pick one or more communication protocols
   - ZigBee, Bluetooth LE, RF433, WiFi (REST, MQTT, CoAP, etc.), …
3. Choose your microcontrollers (Mist)
   - ESP8266, ESP32, Atmel, Nordic, …
   - Sometimes, a Operating System can be used: FreeRTOS, Zephyr,…
4. Get a protocol-compliant gateway (Fog)
   1. Raspberry Pi (or other SBC), Full-fledge server,…
5. Pick a cloud provider
   - AWS, Azure,…
   - Out-of-the-shelf solution: Google Assistant, IFTTT, …

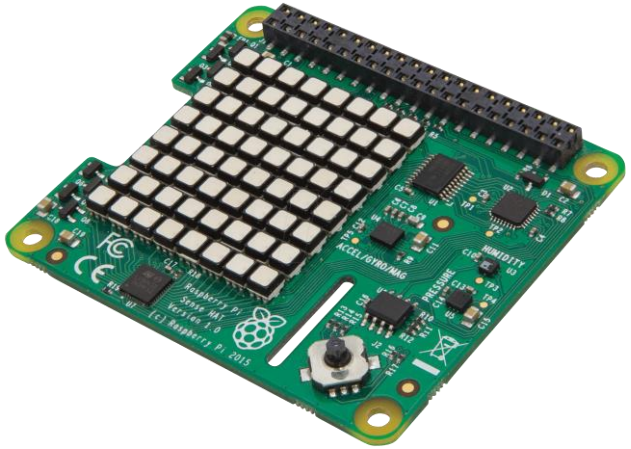
The Software

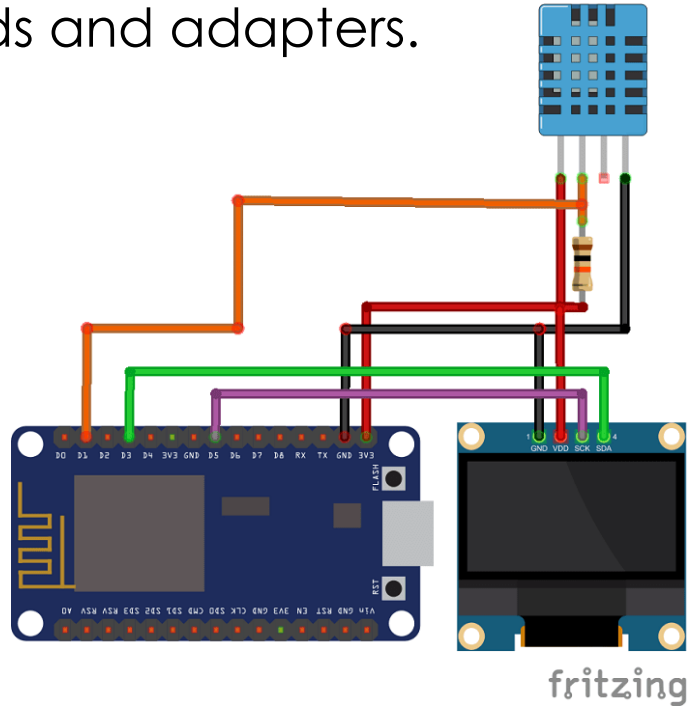
- Edge devices:
  - C, Arduino, microPython
- Fog devices:
  - Full-fledge Linux
  - Node-RED, Domoticz, Home Assistant, OpenHab…
- Cloud:
  - Anything
  - Out-of-the-shelf services

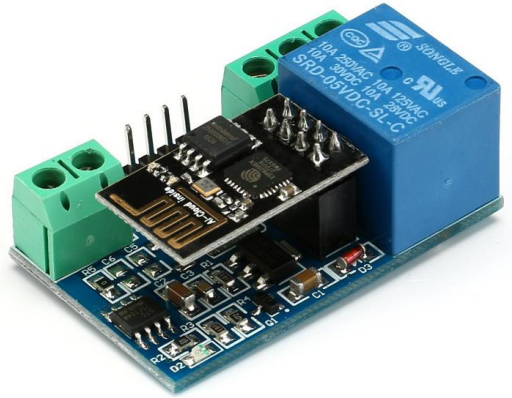
The Hardware

- Flash existent hardware with your software.
  - Serial port is your friend.
- Make your own circuits.
- Buy rapid development boards and adapters.



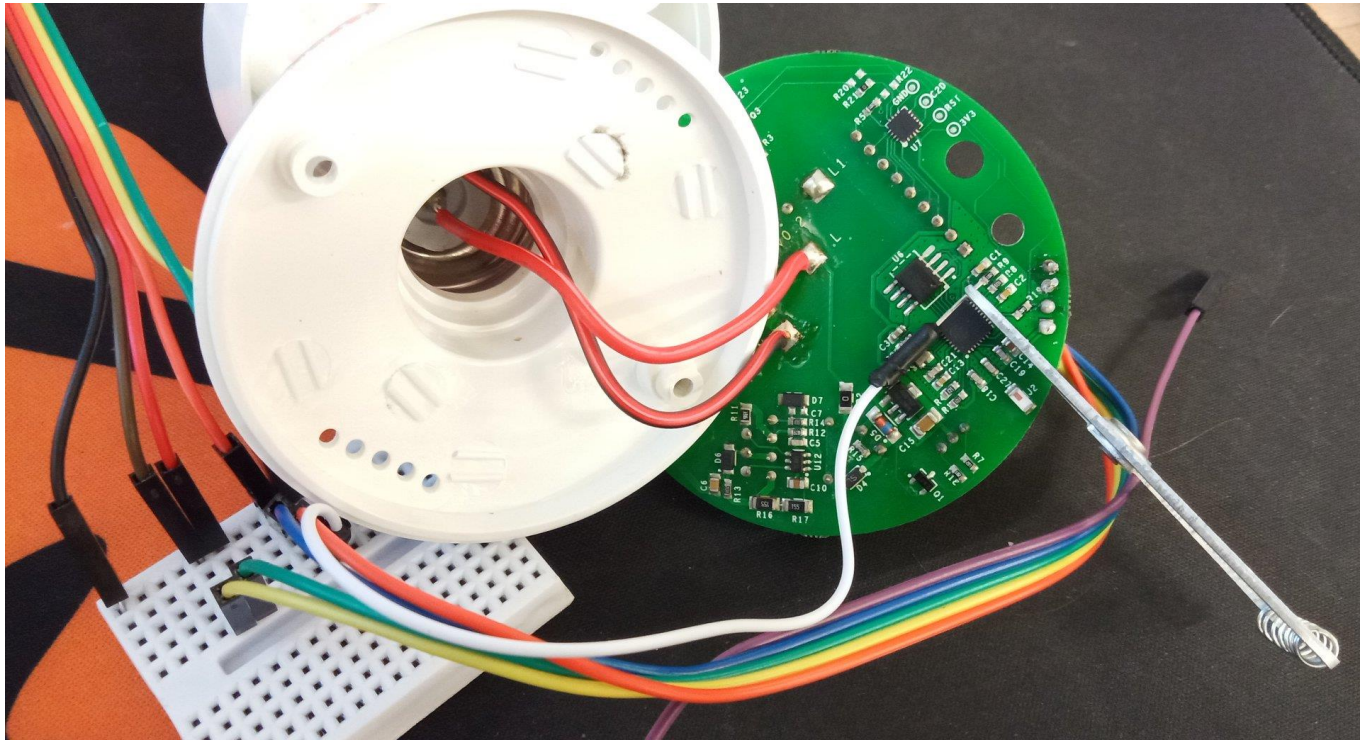






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Flashing a Sonoff Slampher with Tasmota

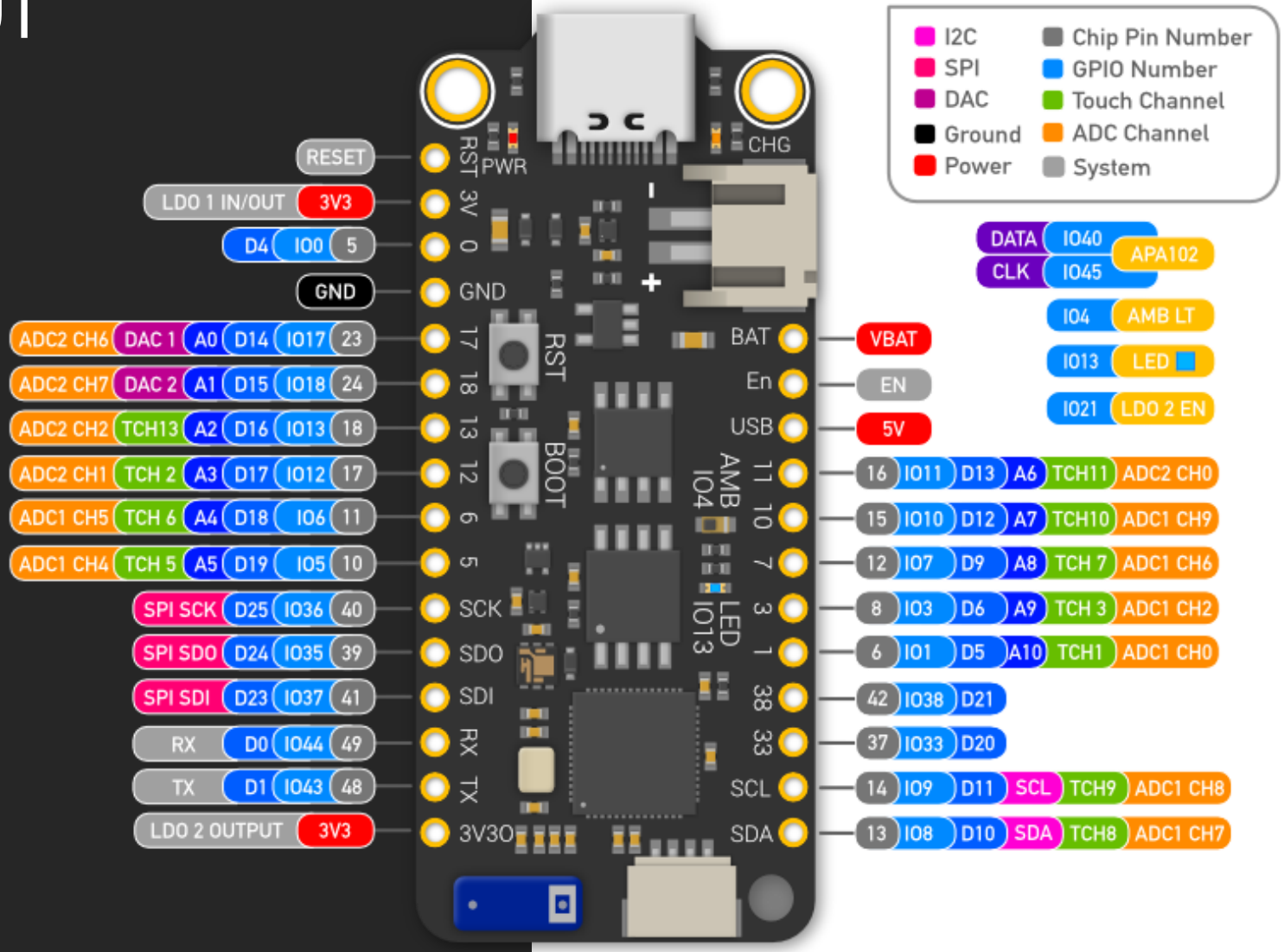




How To: https://tasmota.github.io/docs/devices/Sonoff-Slampher/
Tasmota: https://github.com/arendst/Tasmota

ESP32S2 Pinout

CodeWeek.



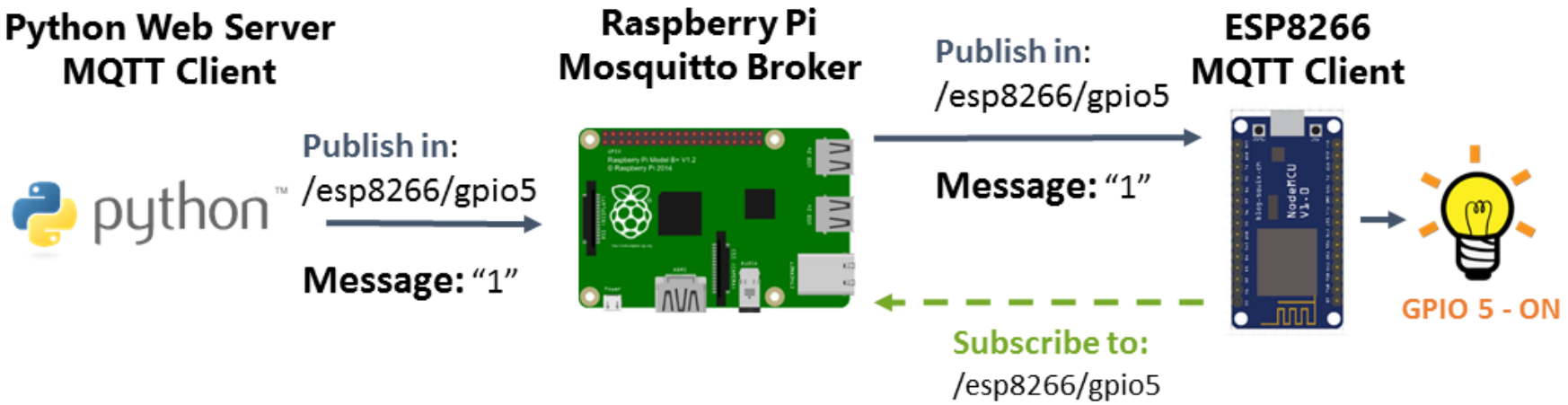
CodeWeek.

I already have some smart things...
Now what?

- Maybe you can *flash* it!
- Welcome to the **protocol dongle jungle.**
  - Make bridges for existent protocols.
  - Typically you will need some specific hardware and software to convert between protocols.
  - This is one of the core features of the **fog** tier.

- Zigbee to MQTT bridge
  - https://www.zigbee2mqtt.io/
- IR blaster
  - https://github.com/mdhiggins/ESP8266-HTTP-IR-Blaster
- RF433, IR, BLE broker
  - https://docs.openmqttgateway.com/

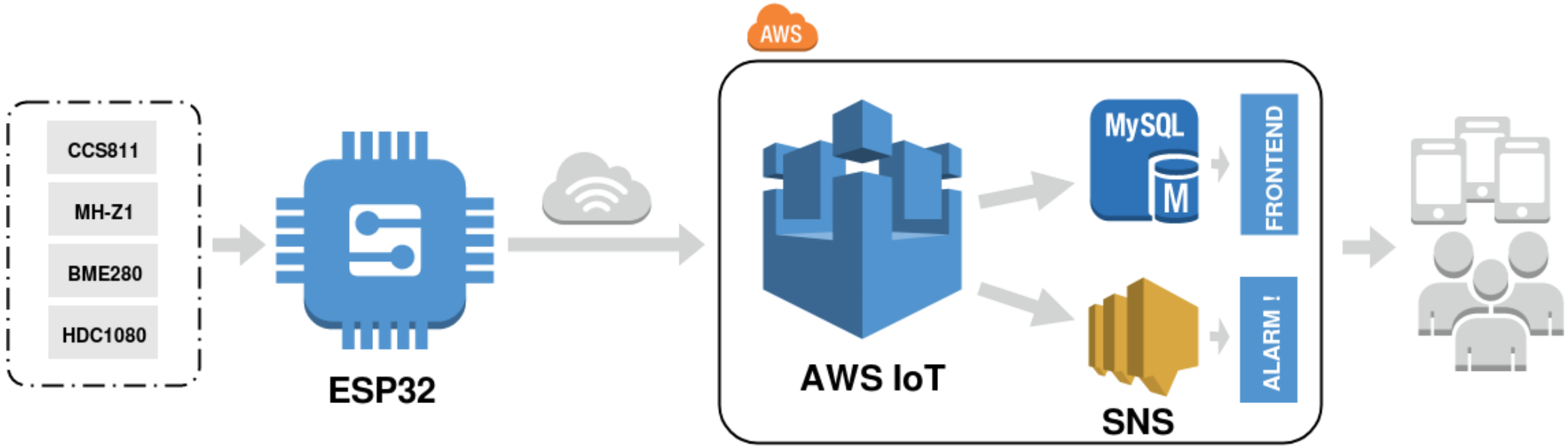
How it all comes together
The MQTT + WebServer Way



Source: https://randomnerdtutorials.com/raspberry-pi-publishing-mqtt-messages-to-esp8266/

How it all comes together

The Cloud All-in



Source: https://sudonull.com/post/3560-Configuring-data-transfer-from-the-device-to-AWS-IoT-Core

Home Assistant example

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Virtual hands-on workshop

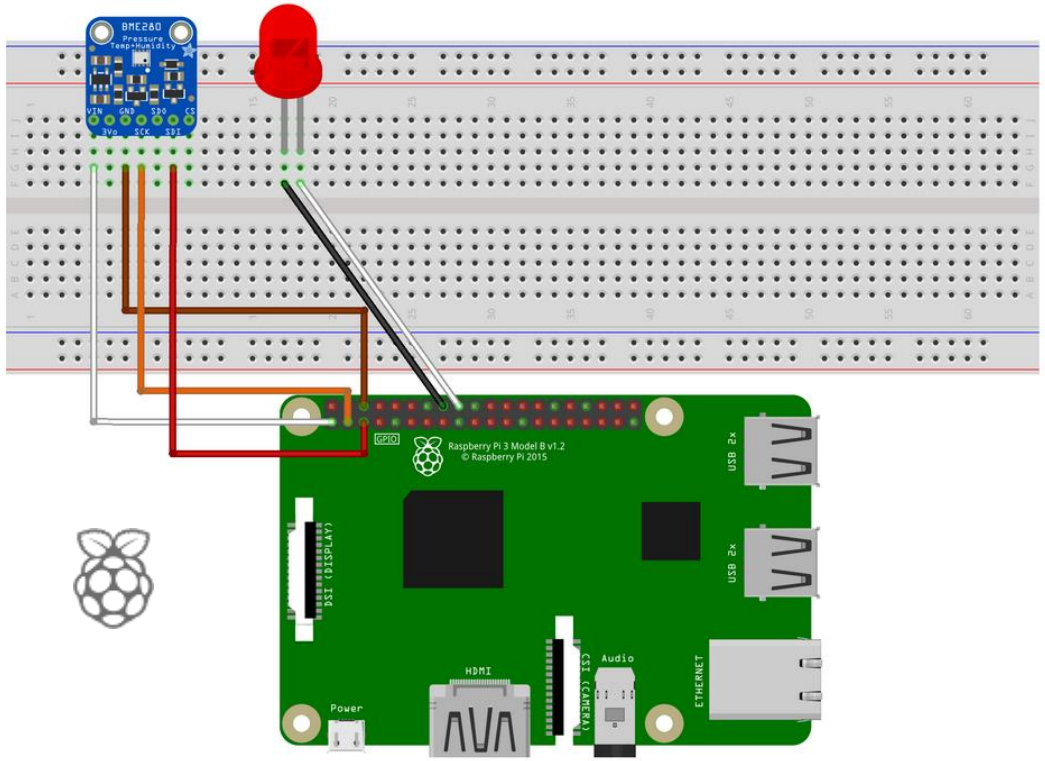
Hammer time!

Goals

1. Toggle a LED.
2. Read data from a sensor.
3. Toggle the LED depending on sensing data.
4. Send sensing data over the web.
5. Request weather data and act upon it.
6. Try different conditions and change things around.

The virtual way

- Raspberry Pi Azure IoT Online Simulator
  - https://azure-samples.github.io/raspberry-pi-web-simulator/
- Coded in JavaScript with WiringPi
  - https://github.com/WiringPi/WiringPi-Node
  - **Delete all the existent code**! It's for Azure related stuff.
- The Raspberry Pi will be our "edge" device
  - But, typically, that's not the case.
- Hardware available:
  - BME280: humidity, barometric pressure and ambient temperature sensor
  - Red LED



CodeWeek.

1. Toggle a LED

```
//Import wiringPi
const wpi = require('wiring-pi');
//Set pin to which the LED is connected
const LEDPin = 4;

//wiringPi setup
wpi.setup('wpi’);

//set LEDPin as an OUPUT (we will change its status)
//set LEDPin default status to off
wpi.pinMode(LEDPin, wpi.OUTPUT);
wpi.digitalWrite(LEDPin, 0);

//write to LEDPin the ON status
//Set the voltage to 5V (or 3.3V on 3.3V boards) for 1 (HIGH), 0V (ground) for 0 (LOW)
wpi.digitalWrite(LEDPin, 1);

//wait for 0.5 seconds and then turn off the LED
blinkLEDTimeout = setTimeout(function () {
  wpi.digitalWrite(LEDPin, 0);
}, 500);
```

2. Read data from a sensor (1/2)

```
//Import wiringPi
const wpi = require('wiring-pi');
//Import sensor lib
const BME280 = require('bme280-sensor');

//wiringPi setup
wpi.setup('wpi');

//device configurations
const BME280_OPTION = {
  i2cBusNo: 1, // defaults to 1
  i2cAddress: BME280.BME280_DEFAULT_I2C_ADDRESS() // defaults to 0x77
 };
```

- What is I2C?
  - Is a synchronous, multi-master, multi-slave, packet switched, single-ended, serial communication bus.
  - https://www.circuitbasics.com/basics-of-the-i2c-communication-protocol/

2. Read data from a sensor (2/2)

```
//instantiate BME sensor
sensor = new BME280(BME280_OPTION);
sensor.init()
 .catch(function (err) {
  console.error(err.message || err);
 });

//Read sensor data and log
sensor.readSensorData().then(function (data) {
  console.log(data)
});
```

- Expected output:


```
{
    "temperature_C":26.090723800037097,
    "humidity":67.46105997902815,
    "pressure_hPa":10.687267861684184
}
```

3. Toggle the LED depending on sensing data

- Based on the previous code:

```
sensor.readSensorData().then(function (data) {
  if(data.humidity > 50){
    wpi.digitalWrite(LEDPin, 1);
  }
});
```

CodeWeek.

4. Send sensing data over the Internet

- Edit the previous code and add a fetch POST request.

```
sensor.readSensorData().then(function (data) {
  console.log(data)
  fetch('https://hookb.in/<provided_during_workshop>', {
    method: 'POST', // or 'PUT'
    mode: 'no-cors',
    headers: {
      'Accept': 'text/plain',
      'Content-Type': 'text/plain'
    },
    body: JSON.stringify(data),
  })
})
```

CodeWeek.

5. Request weather data and act upon it.

```
fetch("https://api.jsonbin.it/bins/el0gfqit")
  .then(resp => resp.json())
  .then(data => {
    if(data.co2ppm > 300){
      wpi.digitalWrite(LEDPin, 1);
    }
  })
```

- The API endpoint is a mock weather data endpoint.
  - In real life, real weather services are used. There are several free.

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Danger Zone

Some recommendations.

Vendor lock-in

- "*vendor lock-in*, also known as proprietary *lock-in* or customer *lock-in*, **makes a customer dependent on a *vendor* for products and services**, **unable to use another** *vendor* **without substantial** switching **costs**. "

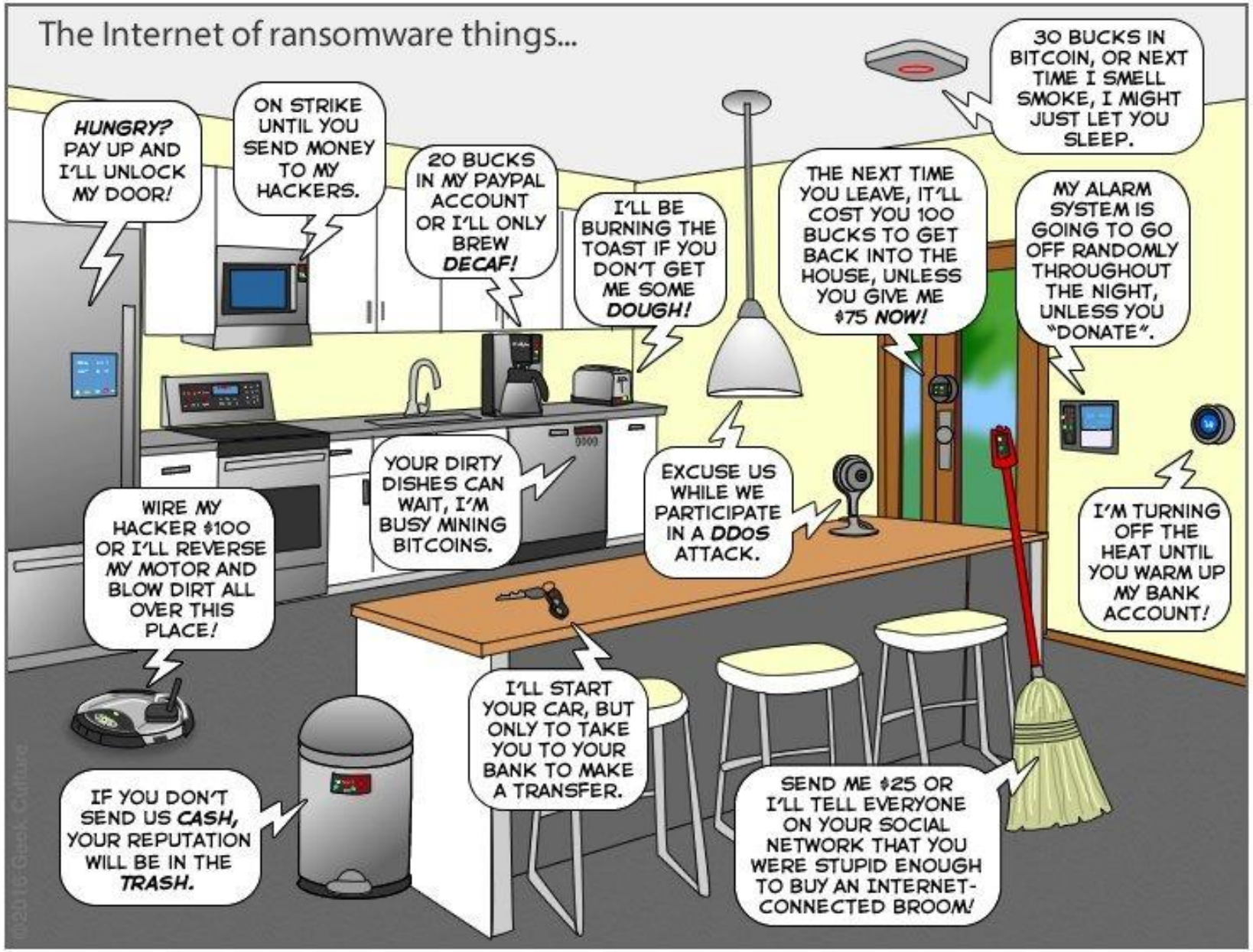
  From Wikipedia, the free encyclopedia

- *Sometimes there are workarounds:*
  - https://github.com/homebridge/**homebridge** for Apple HomeKit
  - Flash, root, and other solutions also exist for some devices.

The **security** side

- IoT systems are more sensible than most software-only things, because **things** can **affect the real-world**.
- Think before you expose your infrastructure over the web.
  - And, when you do it, **do it securely** (e.g. over VPN).
  - Try to not end on **Shodan**: https://2000.shodan.io
- IoT devices are not made to be long-lived.
  - Eventually, vulnerabilities will eventually appear, and no patch will be made.

The Joy of Tech™ by Nitrozac & Snaggy



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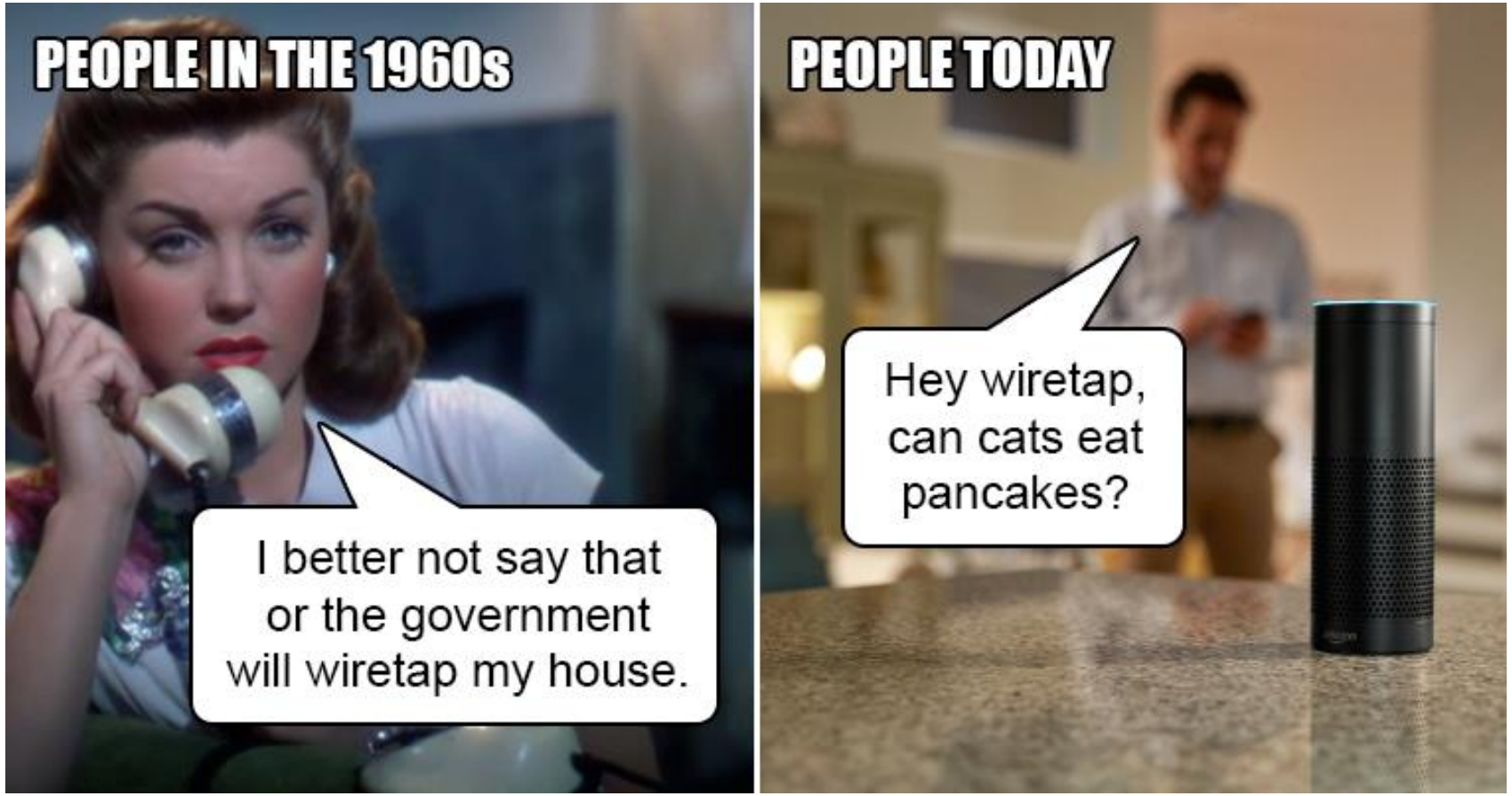
joyoftech.com

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The **privacy** side

- When you buy a device, you can buy more than you want to.
  - Identification
  - Localization and Tracking
  - Profiling
  - Privacy-violating interaction and presentation
  - Lifecycle transition
  - Inventory attack
  - Linkage

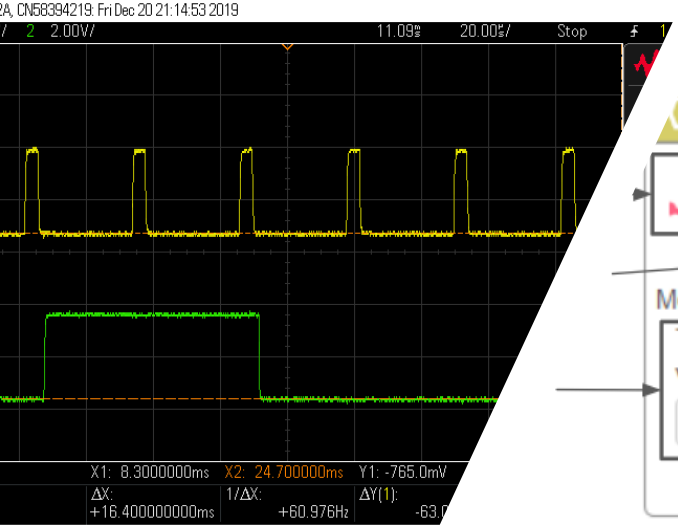
- *Privacy in the Internet of Things: Threats and Challenges*
  - https://arxiv.org/pdf/1505.07683

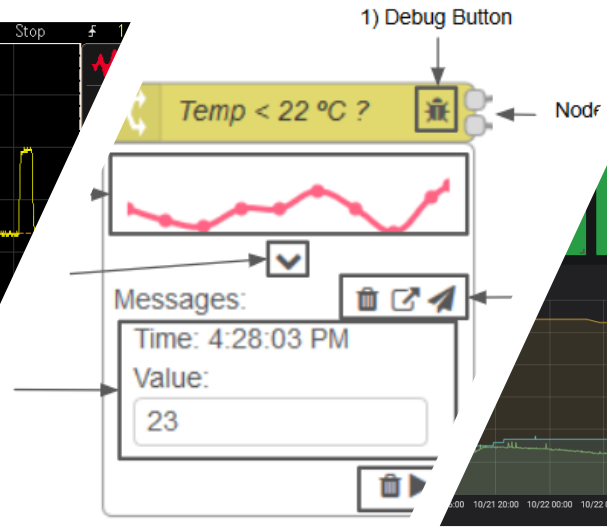

PEOPLE IN THE 1960s
I better not say that or the government will wiretap my house.
PEOPLE TODAY
Hey wiretap, can cats eat pancakes?

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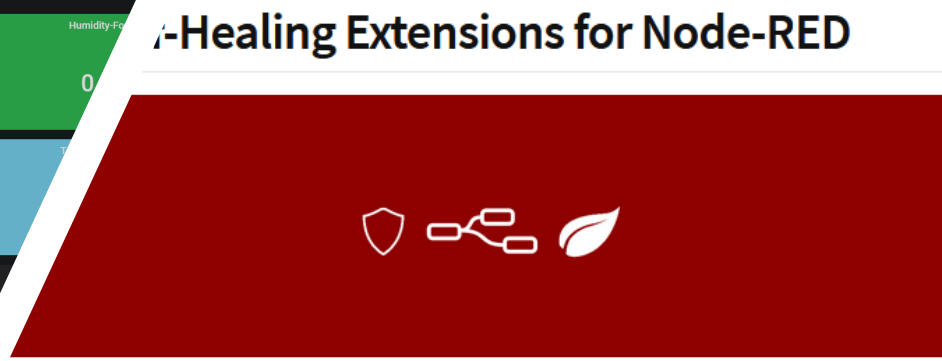
Read it *later*

- An IDE for programmable things: https://platformio.org/
- *The Internet of Risky Things: Trusting the Devices That Surround Us*
  - Book by Sean Smith
- awesome-iot list: https://github.com/HQarroum/awesome-iot
- WebThings for an open standard IoT: https://webthings.io/
- OWASP IoT Project: https://owasp.org/www-project-internet-of-things/
- Fun++:
  - https://twitter.com/internetofshit
  - https://www.shodan.io
  - https://www.iotvillage.org









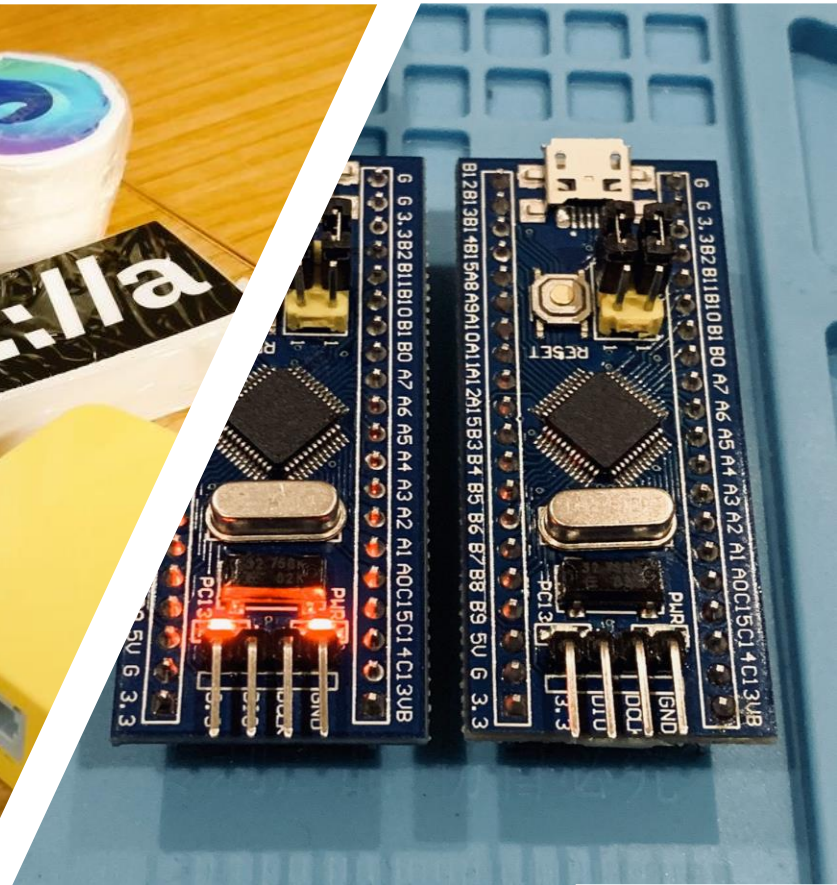
ode-red-contrib-self-healing

DOI 10.1007/978-3-030-50426-7_27 npm v0.4.0 downloads 111/month License MIT

Call for interest

- IoT research lines:
  - Software Engineering
  - Visual programming and low-code
  - Orchestration heterogeneous systems
  - Autonomic Computing (self-healing)
  - Fault-tolerance
  - Privacy and security
  - Embedded and retro computing







Find me @ http://jpdias.me , Twitter and jpmdias@fe.up.pt